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9/28/2025

Unified Health ID: Overcoming Fragmented Records & Low ABHA I'd Adoption

Harshit Rathore, Gauri Pachauri

ABSTRACT

India has an increasing burden of chronic conditions such as diabetes, hypertension, cardiovascular diseases, and cancer that together account for a large share of mortality in the country. To address this issue, initiatives like the Ayushman Bharat Digital Mission (ABDM) and the ABHA health ID have been launched to improve healthcare delivery. Their impact is limited due to persistent challenges such as fragmented medical records, low digital penetration in rural and semi-urban areas, and the lack of integrated frameworks supporting long-term recovery and continuity of care.

These challenges lead to multiple systemic problems. Patients are subjected to repeated tests, delays in treatment, increased costs with no participation in managing their own health. Clinically, doctors do not have access to complete longitudinal patient histories; researchers cannot obtain structured anonymized datasets for meaningful medical analysis. To fill these gaps, this project proposes a patient-centric digital health identification system that will consolidate lifelong medical records starting from birth. The system will provide secure consent-based data sharing and personalized evidence-informed lifestyle and natural care recommendations integrated with conventional treatment pathways. A web-based patient interface and clinician dashboard are proposed here to describe the functionality and usability of the system. The continuity of care, reduction of redundancy in diagnostics and costs, increase in patient engagement through personalized insights as well as reliable data for medical research and policy formulation are expected outcomes. This project will help build resilience by addressing such challenges toward an integrated digital health ecosystem in India.**INDEX TERMS** Digital Health, Non-Communicable Diseases (NCDs), Cancer, Telemedicine, Unified health ID

I. INTRODUCTION

Evidence from national and international health assessments has shown that non-communicable diseases (NCDs) like diabetes, cardiovascular disorders, and cancer are major contributors to premature mortality in India. These diseases do not fall under the category of acute infectious diseases; they are chronic conditions that develop over time and are influenced by multiple determinants such as lifestyle changes, environmental exposure, and population ageing. Such conditions require long-term integrated care strategies due to their nature. Recent surveillance findings and policy discussions have emphasized the need to shift health system priorities from infection control towards sustained management of NCDs in response to this changing disease demography.

- This paper is based on three main pieces of evidence: cancer incidence and prevalence data from the ICMR 2025 registry synthesis covering 43 Population-Based and Hospital-Based Cancer Registries (PBCRs/HBCRs); mortality patterns with trend estimates as reported in global burden of disease studies; programmatic performance documentation from national non-communicable disease programs. When viewed together, these data sources provide age-adjusted incidence indicators, lifetime risk estimates, and program linkage measures underscoring the importance of maintaining longitudinal health records as well as continuity in treatment.

- Mortality Data: This trend analysis is informed by results from the 2025 Lancet study. It was a global assessment that reviewed mortality data across 185 countries for the years between 2001 and 2019. The assessment quantified changes in the probability of death due to non-communicable diseases between birth and 80 years of age with specific reference to India.

- Policy and Performance Data: The government response and programme implementation will be assessed using progress reports for the National Programme for Prevention and Control of Cancer Diabetes Cardiovascular Diseases and Stroke (NPCDCS/NP-NCD) provided by the Ministry of Health and Family Welfare (MoHFW) as of August 2025. These reports give cumulative information about volumes of screening done, outcomes of diagnosis, and linkages to treatment for five priority non-

communicable diseases hypertension, diabetes, oral cancer breast cancer cervix cancer among people aged over 30 years with service delivery mainly through Ayushman Arogya Mandirs (AAMs).

The ICMR analysis also identified the clear pattern of gender-specific patterns in cancer incidence

• Males: The three most frequently diagnosed cancers in the country are oral, lung, and prostate cancers.[4] Oral cancer has become a significant public health issue, with its incidence on the rise in 14 population-based cancer registries (PBCRs) [4][5]. Tobacco-related malignancies continue to represent the majority of the cancer burden among males and are projected to contribute 27.1% of total cases in 2020 [5]. Cancers associated with the gastrointestinal tract also form a large part of this comprehensive disease burden.

• Females: The top sites for cancer are breast, cervical, and ovarian cancers[6].[4] Cervical cancer is on a declining trend in most registries which may reflect the success of vaccination and screening efforts; however, breast cancer incidence rates have significantly increased among many registries.[5] TABLE 1: KEY EPIDEMIOLOGICAL METRICS OF CANCER BURDEN IN INDIA (2025 ICMR ANALYSIS)

Metric	**National Average/Key Finding**	**Extreme Regional Outlier**	**Specific AAIR (per 100,000)**
Lifetime Risk (All Cancer)	11.0%	Mizoram (Males)	21.1%[6] ,[7]
Highest Age-Adjusted Incidence Rate (AAIR)	N/A	Aizawl District (Males)	256.1 (95% CI, 245.2–267.0) [6], [7]
Most Common Cancers (Males)	Oral, Lung, Prostate	N/A	Oral cancer increasing significantly[4]
Most Common Cancers (Females)	Breast, Cervical, Ovarian	N/A	N/A [6],[4]

Comparative Analysis of India's Mortality Ratio

One of the most evident signs of structural weaknesses in the cancer control system in India is the high Cancer Mortality-to-Incidence Ratio (MIR) that has been reported. Though incidence figures tend to attract more public attention, MIR reflects more directly what a health system can do to ensure timely diagnosis and treatment, as well as continuing care that affects survival. India's cancer mortality-to-incidence ratio was reported at 64.47% for 2022—this is one of the highest ratios ever reported among countries with large absolute burdens of cancer [12]. Such a high ratio indicates that there are structural problems in turning diagnosed cases into good long-term treatment results rather than problems simply limited to detection of disease.

This is sharply at variance with what has been achieved by higher-income health systems. The United States has an MIR of just 23.81% though it has a greater absolute number of cases; India's MIR is worse than those for several other emerging economies—in addition to China with 50.57%, and Indonesia with 58.66% [8].

Quantifying India's Outlier Status

In 2025, The Lancet published a study that highlighted the clear difference between India and the rest of the world in controlling non-communicable diseases (NCDs) [1][2]. At the global level from 2010 to 2019, almost four out of five countries experienced a reduction in the chance of dying prematurely from chronic diseases. During this same period, India saw an increase in the probability of premature deaths due to NCDs [1][2].

The Longitudinal Ageing Study in India (LASI) also throws some more light on how unevenly NCD-related mortality is spread across different socioeconomic groups. Among adults aged 45 years and above, the share of deaths due to NCDs is highest within lower-income groups at 43.08%, followed by the highest income group at 39.54% [9]. This shows that NCDs are now no longer just diseases of affluence; while lifestyle-related risk factors may be more prevalent among higher-income populations, economically disadvantaged groups have increased

mortality risks due to late diagnosis, breaks in long-term treatment, and inadequate financial resources to deal with chronic illnesses. Vulnerable and socially marginalized populations thus bear disproportionately higher morbidity and mortality burdens without consistent access to comprehensive healthcare services [10].

Policy Implementation and Program Performance: The NPCDCS Review (2024-2025) The Ayushman Bharat Digital Mission (ABDM) along with the ABHA ID is an important policy initiative that aims at creating a unified paperless healthcare ecosystem. The actual impact of this initiative depends on public awareness and user adoption levels. Survey responses collected from 56 young adults in the age group of 16 to 25 years reflect early-stage understanding and perception regarding this digital health framework among key user demographics.

TABLE 2: NPCDCS PROGRESS ON SCREENING AND TREATMENT (TARGET POPULATION: AGE:30+), AS OF AUGUST 2025 [11]

Indicator	Hypertension (in crores)	Diabetes (in crores)	Oral Cancer (in lakhs)	Breast Cancer (in thousands)	Cervical Cancer (in thousands)
Screening Conducted	38.96	38.77	32.81	17.21	10.21
Diagnosed Cases	5.13	3.46	1.88	41.39	90.29
Cases Under Treatment	4.75	2.84	1.77	61.74	97.09

Awareness Gap: The survey results reveal a significant gap in digital health literacy among young adults. Only 37.50% of respondents were fully aware of the purpose and function of the ABHA ID, while the majority, 46.40%, had partial knowledge; 16.04% were completely unaware. This gap indicates that despite large-scale national initiatives, the communication strategy about the mechanics and utility of digital health accounts is not effectively reaching the youth, who are often perceived as early adopters of technology.

The survey indicates that awareness among Indian youth about this issue is very low.

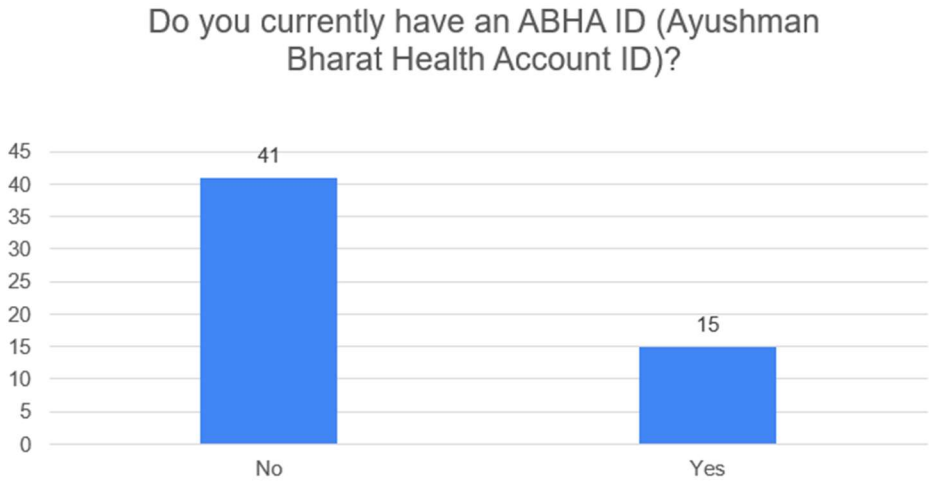

FIGURE 1: - Survey responses on whether participants have an ABHA ID

Only 15 out of 56 have an ABHA I'd that is about 25%; it is depressing to see least awareness amongst the youth.

To understand whether the youth have a little knowledge about this cause and the need of such an initiative another survey question and its stats are as follows:

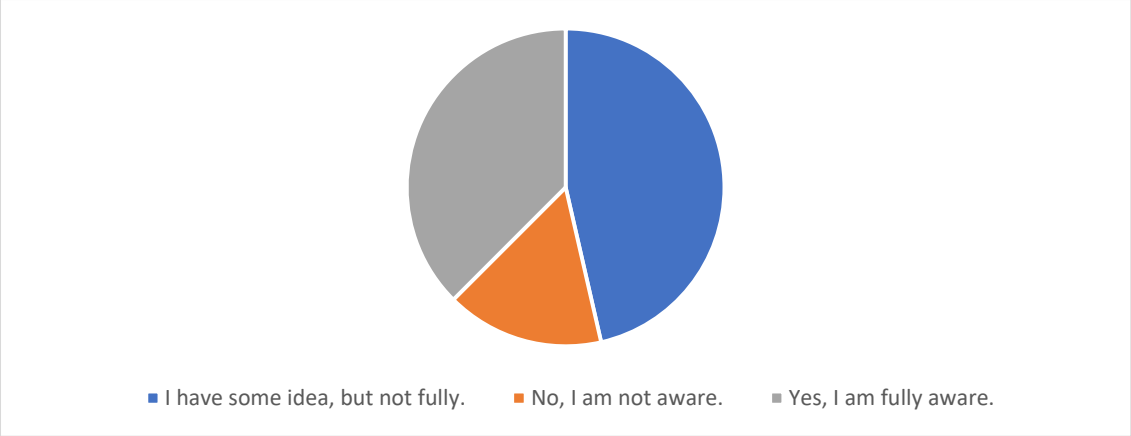

FIGURE 2: - ABHA ID awareness levels among young adults.

TABLE 3: ABHA ID AWARENESS AMONG YOUNG ADULTS (AGE 16-25, N=56)

Are you aware of what an ABHA ID is and its purpose?	**Numbers**	**Percentage**
I have some idea, but not fully.	26	46.40
No, I am not aware.	9	16.04
Yes, I am fully aware.	21	37.50

Taking into consideration the above statistics from national and international organizations, as well as the recent survey conducted during the research, it shows that there is a need for synchronized, consecutive, and organized data to prevent, regulate, and monitor the health trades of India. This enables a comprehensive way to study, store, and use the health records of patients.

Storing medical history from an early age (i.e., at birth) allows patients to retrieve and organize their data in a way that is easily accessible to both themselves and doctors when needed. This approach not only reduces patients' time in finding and presenting medical reports but also replaces paperwork to a much fast and easily accessible way. Keeping in mind the need to study the health of our country's health for further research, this data can help build a strong foundation for such efforts. The relevance of this system is seen in chronic disease management. Access to complete and accurate medical histories is paramount for treatment decisions. In many instances, patients from remote or underserved regions are unable to provide comprehensive health information, resulting in treatment delays, complications related to medication, and less than optimal recovery outcomes. These issues further underline the necessity for a national health identification system that is universally accepted. Within that framework, lifelong records integrated into the system akin to mandatory academic identification systems would allow for enhanced continuity of care. Therefore, this project develops a unified digital portal aimed at supporting coordinated access by patients, healthcare providers as well as government stakeholders.

II. METHODOLOGY: DESIGNING A UNIFIED, LONGITUDINAL DIGITAL HEALTH IDENTIFICATION SYSTEM

The major limitations discovered in the preliminary analysis were high cancer Mortality-to-Incidence Ratio (MIR) [8], fragmented health records, and low public uptake of existing digital health initiatives. This describes a need for an integrated and patient-centered digital health infrastructure. Hence, this paper presents a proposal to design and develop a comprehensive three-tier web-based application that aims to keep continuous longitudinal health data records for individuals starting from birth.

A. Architectural Foundation: Unified Health Identifier and Lifelong Health Data Framework The proposed methodology is based on the issuance of a Unified Health Identifier at birth so that it would be possible to create a continuous lifelong health record. Early assignment of the identifier enables immunization, neonatal, and

growth-related data to be captured in one longitudinal framework thus reducing fragmentation of data from inception. This is in line with existing guidance regarding national health identifiers which are recognized as key components necessary for continuity of care at the population level as well as being consistent with internationally accepted practices regarding persistent patient identification.

B. Component 1: The Patient Portal

The Patient Portal is a secure web-based application designed to allow users to view and manage their personal health information.

• Access and Authentication: The portal will be used by patients with their unique UHI and a verified password. Appropriate security mechanisms will be used to allow safe, consent-based access to personal medical data.

• Functionality: The portal allows patients to view and organize their full medical history, including physician prescriptions and diagnostic reports. Direct access to these records helps reduce administrative delays as well as the unnecessary costs of duplicate diagnostic testing when patients transfer between healthcare facilities or geographical areas.

C. Component 2: The Clinician Dashboard

The Clinician Dashboard is a one-stop view for clinicians, enabling them to make better decisions and ensure that care for patients is continuous.

• Use: Once authenticated securely with verified credentials, clinicians can access, based on consent, the entire longitudinal medical record of a patient.

1. Data Entry: Doctors can enter prescriptions and clinical notes along with other health data directly into the patient's record linked to UHI; thus, keeping this digital file always updated.
2. Clinical Effectiveness: Instant access to all past medical information about a patient - previous drugs taken by the patient, current chronic conditions of the patient, and recorded family history - aids in making right decisions regarding prescribing medicines. It lowers chances of complications due to medicine and helps find possible risks from inherited diseases. This long view is very key when handling management for non-communicable diseases (NCDs).

D. Component 3: The Central Administration and Surveillance Panel

The panel acts as the governance layer of the digital ecosystem, providing assurance of data integrity, regulatory compliance, and national health surveillance.

• Features:

1. Data Store: It serves as a secure store for all authenticated records submitted via the Patient Portal and Clinician Dashboard.

2. Clinician Authentication and Audit: The panel keeps records of all registered clinicians, checks their credentials, and logs all permitted access to and updates of data to enable professional accountability and secure data management. 3.

Public Health Surveillance: Through the panel, authorized organizations and researchers are able to access anonymized and aggregated datasets from the longitudinal health database under strict security controls. These datasets are used to study trends in national health, assess major diseases, and predict new public-health threats (e.g., early signs of a pandemic). This reinforces the evidence base for further research and policy-making.

This integrated system enables a coordinated, sequential, and structured exchange of information that replaces traditional paper-based approaches with an instantaneously available digital solution supporting patients, clinicians, and public health decision-makers.

III. RESULTS

A review of recent epidemiological data (2019-2025) and national program performance reveals a stark contrast in India's health landscape: impressive disease screening volumes against alarming mortality rates and significant implementation gaps at the system level.

A. Worsening NCD and Cancer Outcome Indicators

- Global Mortality Difference: India stands out as a global health anomaly by showing an increase in the probability of dying prematurely from non-communicable diseases (NCDs) between 2010 and 2019, unlike other countries where this trend has been decreasing. This unfavorable trend has primarily been driven by heart disease and diabetes conditions.
- Gender Risk Disparity: The increase in NCD mortality probability was substantially worse for women, with an increase of 2.1 per cent, while men's increase was marginal at about 0.1 per cent.
- Cancer Survival Indicator: India's cancer Mortality-to-Incidence Ratio (MIR) recorded at 64.47% in the year 2022 is the best indicator of ultimate failure in outcome since it has the highest ratio among ten countries with a maximum burden of cancers which means that almost two-thirds of newly diagnosed patients do not survive the disease.

B. NPCDCS Performance: Volume versus Linkage

The NP-NCD program has achieved good coverage regarding screening but shows a marked drop in follow-up and management as per data presented in MoHFW's progress report for August 2025.

TABLE 2: NPCDCS Progress in Screening and Treatment (Target Population: Age ≥30+), AS OF AUGUST 2025 [3]

Indicator	**Hypertension (in crores)**	**Diabetes (in crores)**	**Oral Cancer (in lakhs)**	**Breast Cancer (in thousands)**	**Cervical Cancer (in thousands)**
Screening Conducted	38.96	38.77	32.81	17.21	10.21
Diagnosed Cases	5.13	3.46	1.88	41.39	90.29
Cases Under Treatment	4.75	2.84	1.77	61.74	97.09

Program data show fairly strong linkage to care for some conditions, like hypertension and oral cancer. However, there is a major gap for diabetes: about 18 percent of those diagnosed—around 0.62 crore people—are not in ongoing follow-up and treatment [3]. This points to weaknesses in chronic care pathways and in systems that track and support long-term adherence by patients.

C. Digital Health Adoption and Awareness Gap

Survey data from young adults aged 16 to 25 reveal a notable inconsistency in the adoption of digital health initiatives:

- High Optimism: A combined 73.21% of respondents either 'Agree' (35.71%) or 'Strongly Agree' (37.50%) that the ABHA ID will improve healthcare access and services.

- Limited Knowledge: Though there was a favorable anticipation, only 37.5% of those surveyed could claim complete comprehension of the ABHA ID's intent and operation, with the most significant group (46.4%) having only partial awareness.

This gap indicates that while there is overall public receptivity to a digital health ecosystem, the current communication and implementation strategies have not successfully built the foundational level of digital health literacy required for broad adoption and effective use.

TABLE 3: PUBLIC OPINION ON ABHA ID IMPROVING HEALTHCARE ACCESS (AGE 16-25, N=56)

Opinion	Numbers	Percentage
Agree	20	35.71
Neutral	15	26.78
Strongly Agree	21	37.50

TABLE 4: ABHA ID AWARENESS AMONG YOUNG ADULTS (AGE 16-25, N=56)

Are you aware of what an ABHA ID is and its purpose?	Numbers	Percentage
I have some idea, but not fully.	26	46.40
No, I am not aware.	9	16.04
Yes, I am fully aware.	21	37.50

IV. Key Findings

Key findings from the integrated analysis of epidemiological data, government documents, and primary survey results are listed below:

- Failure in Survival Outcomes: India has the highest cancer Mortality-to-Incidence Ratio at 64.47%, representing systemic failures to convert early diagnosis into continuing life-saving treatment.
- Global Mortality Divergence: India is an outlier in NCD management with increasing probabilities of premature NCD-related deaths between 2010 and 2019, primarily due to heart disease and diabetes.
- Gender and Socio-Economic Inequity: The increase in NCD mortality more greatly affects women (2.1% increase) and lower-income groups, revealing substantial financial and geographical barriers to care.
- Programmatic Linkage Gap: The NPCDCS, despite achieving high screening volumes, suffers from a severe linkage failure, evidenced by the approximately 18% of diagnosed diabetes cases not under sustained treatment
- Digital Literacy Challenge: Despite the high public confidence in digital health (73.21% optimism regarding the ABHA ID), actual digital literacy and awareness are limited, with only 37.5% fully informed, creating a significant barrier to effective implementation.

V. DISCUSSION

The results show that there is a big gap between what India is doing in public health and what is needed, as seen in the NP-NCD program. Even though a lot of screenings have been done [3], a drop in illness and death from NCDs was expected but has not happened [1][8]. This problem happens because of three main issues: difficulties with putting plans into action, ongoing social and economic inequalities between genders, and important shortcomings in digital health systems.

A. Systemic Failure in Treatment Linkage

- The high cancer Mortality-to-Incidence Ratio of 64.47% [8] indicates systemic failures post-diagnosis, with links to widespread operational challenges at both the micro and meso levels of the NP-NCD framework. The high cancer MIR and diabetes linkage gap are indicative of failures occurring post-diagnosis, specifically operational and resource constraints at facility and programmatic levels [12]. Unfilled clinical positions, intermittent availability of essential medications, and inadequate follow-up mechanisms lead to disrupted care pathways that undermine patient adherence and increase the risk of mortality [15][16][13]. Therefore, addressing these implementation bottlenecks [3] is critical for translating screening successes into improved survival outcomes.

- Surveillance Crisis: Effective national policy implementation suffers from the absence of a fully implemented monitoring framework for the NPCDCS [12][13][14] plus low-quality health data [15]. This “data deficit” makes it impossible to quickly find bottlenecks and also leads to inaccurate reporting [14]. Technical limitations such as bad internet connectivity in remote areas and not user-friendly software further reduce how well current digital reporting tools work. This limits the system's ability to gather structured information about NCD risk factors even though there are national efforts to increase surveillance coverage.

B. Socio-Economic and Gender Disparities

The rise in NCD deaths is mainly attributed to social inequities. The greater chance of death from NCDs among females and the fact that mortality is highest in low-income groups show that these are not only "diseases of the rich."

- Poverty and Treatment Adherence: The impact of NCDs is intensified by limited financial capacity for continuous chronic disease management [17][18]. Individuals from Scheduled Caste (SC) and Scheduled Tribe (ST) communities experience substantially higher risks of non-NCD-related mortality, highlighting persistent social and health inequities.[9]

- The large gap in female NCD mortality shows systemic barriers, including geographical access issues like far-off healthcare facilities and financial constraints that stop 68% of women from getting timely and full non-maternal healthcare.

- Regional and Behavioural Factors: The pronounced regional disparity in cancer incidence, such as the high risk in Mizoram [7], is tied to unique behavioural factors like the consumption of 'Taibu' (tobacco smoke-infused water).[21] Behavioural risk factors drive the highest percentage of Disability-Adjusted Life Years (DALYs) in India (23.95% DALYs) , [22],[23] This highlights the inadequacy of a uniform national intervention strategy, which must be tailored to address specific local cultural and behavioural risk factors.[21]

C. The Necessity of a Unified Digital System

The digital health platform described in the Methodology seeks to address the specific gaps in data interoperability, patient de-duplication, and accessibility in rural areas as noted above [24] [25][26].

- Bridging the Knowledge Gap: High optimism with low digital literacy from the surveys reinforces a need for a mandatory integrated Unified Health Identifier (UHI) at birth to solve issues of low voluntary uptake and technical complexity [27] associated with current initiatives. Family engagement and support by local Primary Health Centre(PHC) practitioners are essential for promoting adoption of such solutions.

- Solving the Fragmentation Crisis: This three-pronged approach will create a longitudinal health record from birth and thus solve fragmentation that now prevails against effective chronic disease management by denying clinicians access to full patient histories, which is necessary for clinical effectiveness and preventing avoidable costs.

- Enabling Surveillance: The Central Administration Panel can thus respond to existing NPCDCS surveillance challenges by safely pooling anonymized data into one repository [12],[28], so that policymakers have access to up-to-date comprehensive datasets about results such as MIR and what evidence-based interventions should be applied for systemic and regional health inequities.

VI. FUTURE SCOPE AND DEVELOPMENT

Over the next five decades, India’s healthcare sector is expected to shift toward predictive, personalized, and decentralized care, with strong dependence on reliable digital infrastructure. The proposed Unified Health Identifier (UHI) system provides a vital foundation to facilitate this transition and to promote equitable access, including for the most vulnerable populations.

A. Enhancing Inclusivity: Tackling Digital Literacy Gaps and Bridging the Rural Divid

- The system's future development must take into account the digital divide and digital anxiety, particularly for old and low-literacy populations. Non-Technical Interfaces (Near Term): The system will use the current network of Accredited Social Health Activists (ASHAs) and Auxiliary Nurse Midwives (ANMs) to interface with citizens who have low digital literacy. These frontline workers will be the main interface through the Clinician Dashboard (Component 2), using voice commands and vernacular-language tools to register data and retrieve patient information. Family members will also provide social support for adoption of the system. Telehealth and Infrastructure (Mid Term): The system should integrate advanced telemedicine functionalities, supported by targeted national investments in digital infrastructure and sustained public awareness initiatives to enhance digital literacy, in order to bring tertiary care closer to remote areas. This decentralized model contributes towards overcoming geographical barriers that disproportionately affect women. Voice and Biometric Authentication (Long Term): The UHI system should evolve to accommodate standardized voice and regional biometric authentication, thereby diminishing dependence on complex passwords or language-specific interfaces for accessing vital health records.

B. Research Trajectories for the Next Fifty Years

The creation of an enormous longitudinal UHI-linked dataset opens up, for the first time, research opportunities that were not available before:

- Longitudinal health records consolidated into one dataset provide a strong basis for predictive modelling and risk stratification. When cohorts are carefully selected and governed ethically, such datasets can be used to inform machine-learning models about future disease risks to enable early targeted preventive interventions, although thorough validation and equity safeguards are necessary before clinical implementation.

- Public Health Surveillance and Pandemic Prediction: The Central Administration Panel (Component 3) will provide continuous high-quality data that can be used for modelling disease transmission, identifying new regional NCD hotspots, and creating early warning systems for public health emergencies in the future. This will address the current low quality of health data in India.

- Health Economics and Policy Evaluation: Integrated treatment adherence data plus longitudinal morbidity records allow policymakers to more accurately assess the cost-effectiveness of different interventions by looking at real health outcomes rather than just screening metrics supporting optimal allocation of resources.

C. Future Development of the Unified Health System

Future development of the project should prioritize integration of the system with the expanding medical technology ecosystem:

• Clinical Decision Support Systems (CDSS): Embedding artificial intelligence (AI) within the Clinician Dashboard will deliver real-time, evidence-based guidance for diagnosis and treatment, supporting physicians in underserved regions where access to specialist consultation is limited.

• Wearable Technology Integration: The UHI should implement open and secure APIs to incorporate data from patient-owned wearable devices and remote monitoring tools, facilitating passive tracking of chronic conditions (e.g., blood glucose, blood pressure) and providing patients with personalized, evidence-based lifestyle recommendations (Component 1) aligned with Ayurvedic, Yoga, and Naturopathy (AYUSH) principles.

The UHI system is thus more than a digitization initiative; it represents a national platform aimed at enabling equitable, advanced health management across India's entire socio-economic spectrum over the next fifty years.

VII. CONCLUSION

India's epidemiological transition has reached a tipping point, as reflected in the alarming increase in NCD mortality [1] and a cancer Mortality-to-Incidence Ratio of 64.47%, which is one of the highest ratios globally [8]. This crisis is not due to poor screening [3], but rather reflects major gaps in patient linkage, monitoring, and

continuity of care that are due both to operational issues such as staff vacancies and irregular drug supplies [22], as well as persistent socio-economic inequities.[9]

Effectively addressing the chronic disease epidemic will also require changing the fragmented approach with a strong outcome-oriented digital health ecosystem that meets national health program goals. The proposal for this project includes creating a Unified Health ID system with integrated Patient and Clinician Portals for:

• A universal longitudinal data record from birth.[18]

• Continuous informed care, thus reducing the cancer MIR and improving treatment adherence (especially for conditions like diabetes).

• Accurate real-time data to policymakers for national health surveillance so that timely targeted interventions can be made against systemic and regional health disparities.

The digital health system outlined here is therefore not just necessary but strategic if India wishes to move from tracking disease incidence toward saving lives in its pursuit of sustainable development goals.

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